

PRODUCT REQUIREMENTS DOCUMENT

Internship Management System

IMS — Version 1.0 | MVP Edition

A lightweight web-based platform for managing 20-25
interns
inside a modern startup environment

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Prepared for internal use only

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1. Product Overview

The Internship Management System (IMS) is a lightweight, modern, web-based platform purpose-built for startups that run structured internship programs. It replaces ad-hoc spreadsheets, WhatsApp groups, and manual document generation with a clean, role-aware digital workspace that makes interns feel like they are working inside a real company from day one.

1.1 Vision

"Build the simplest, most developer-friendly internship operating system that makes every intern feel valued, tracked, and equipped — without the complexity or cost of enterprise HR software."

1.2 Objectives

- Digitize the full internship lifecycle — from onboarding to certificate generation.
- Give interns a professional dashboard experience that mirrors a real workplace.
- Empower admins to manage attendance, tutorials, and documents from a single panel.
- Reduce manual effort in offer letter and certificate creation to near-zero.
- Keep the codebase lightweight, maintainable, and deployable under zero infra cost.

1.3 Problem Statement

Most startups managing intern cohorts face a fragmented workflow:

- Attendance is tracked in Google Sheets or WhatsApp, leading to inconsistencies.
- Tutorial resources are scattered across Google Drive, Notion, or email threads.
- Offer letters and certificates are generated manually in Word — time-consuming and error-prone.
- Interns have no centralized dashboard, creating a disconnected, un-professional experience.
- Admins have no real-time visibility into attendance trends or learning progress.

The IMS solves all of these problems in a single, cohesive platform.

1.4 Target Users

User Group	Description
Admin	Startup founder, HR lead, or operations manager overseeing the internship program
Team Lead	Department mentor or senior intern who manages a sub-team of 4-6 interns
Intern	College student (undergrad / postgrad) enrolled in a structured internship program
Future	Clients or external stakeholders who may verify certificates via QR code

1.5 Expected Outcomes

- 100% digital attendance tracking with zero manual intervention.
- Structured learning path for every intern through the tutorial system.
- Automated offer letter and certificate generation saving 3-4 hours per cohort.
- Professional intern onboarding experience from day one.
- Admin visibility into real-time attendance, completion rates, and cohort health.

2. User Roles & Access Control

The IMS uses a three-tier role-based access control (RBAC) model. Roles are assigned at account creation by the Admin and stored in the Firestore users collection. Firebase custom claims mirror the role for server-side enforcement.

2.1 Role Overview

Attribute	Admin	Team Lead	Intern
Scope	Full system access	Own team + own profile	Own profile only
Create Users	Yes	No	No
Manage Attendance	All interns	Own team only	Self only
Upload Tutorials	Yes	Yes (pending admin review)	No
Generate Documents	Yes	No	Download own only
View Analytics	All	Team only	Personal only
Manage Announcements	Yes	Yes	Read only
Approve Certificates	Yes	Recommend only	No

2.2 Admin Role

The Admin is the super-user of the platform. There is typically one Admin per startup, though the system supports multiple Admin accounts.

- Full CRUD access to all user accounts, attendance records, tutorials, and documents.
- Can create intern and team lead accounts and assign roles.
- Has access to system-wide analytics and export functionality.
- Sole authority to approve and publish completion certificates.
- Can configure system settings such as startup branding (logo, name, colors).

2.3 Team Lead Role

Team Leads are senior interns or department mentors assigned to manage a sub-group of interns.

- Can view and monitor attendance for interns in their assigned team.
- Can upload learning materials (subject to Admin approval before publishing).
- Can post announcements scoped to their team.
- Can recommend certificate approval for their team members.
- Cannot access other teams' data or generate documents.

2.4 Intern Role

Interns are the primary end-users of the platform. The entire UX is optimized for their daily experience.

- Can check in and check out to log attendance.
- Can view assigned tutorials and mark them as completed.
- Can view personal attendance history and completion percentage.
- Can download their own offer letter and completion certificate (once approved).
- Can view announcements posted by Admin or their Team Lead.

3. Complete Feature Breakdown

The IMS is composed of seven core modules. Each module is described below with its features, priority, and implementation notes.

Module A: Authentication System

Secure, role-aware authentication using Firebase Auth with email/password and optional Google OAuth.

Feature	Description	Priority
Email/Password Login	Interns and Admins log in with registered credentials. Passwords are handled entirely by Firebase Auth.	P0
Signup Flow	Admin-initiated account creation. Interns do not self-register. Admin creates accounts and shares credentials.	P0
Role-Based Routing	On login, the user's Firestore role document is read and they are redirected to the appropriate dashboard.	P0
Google OAuth (Optional)	Optional Google Sign-In for faster onboarding. Role is still assigned by Admin post-signup.	P2
Password Reset	Firebase built-in password reset via email link.	P1
Session Persistence	Firebase session tokens persist across browser refreshes. Auto-logout after 24 hours of inactivity.	P1
Route Guards	Protected routes check Firebase auth state and Firestore role before rendering any page.	P0

Implementation Note:

Use Firebase Authentication v9 modular SDK. Store the user role ('admin', 'teamlead', 'intern') in Firestore under `/users/{uid}/role`. Use React Context or Zustand to expose the auth state and role globally. Never trust the client-side role alone — validate with Firestore Security Rules on all writes.

Module B: Dashboard System

B1 — Intern Dashboard

Feature	Description	Priority
Attendance Summary Card	Shows today's check-in/out time, this week's attendance count, and overall attendance percentage.	P0
Assigned Tutorials Card	Displays the count of total tutorials, completed tutorials, and a progress bar.	P0
Announcements Feed	Latest 5 announcements from Admin and Team Lead, sorted by date descending.	P0
Profile Overview	Avatar, name, department, internship duration, and Team Lead name.	P1

Quick Actions	One-click Check-In / Check-Out button. Active state reflects current status.	P0
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B2 — Admin Dashboard

Feature	Description	Priority
Total Interns Card	Live count of active interns with a sparkline trend for the past 30 days.	P0
Attendance Statistics	Today's present/absent count, weekly heatmap, and per-intern attendance table.	P0
Tutorials Uploaded	Count of total tutorials by category with a donut chart breakdown.	P1
Certificate Stats	Number of pending approval requests, approved, and downloaded certificates.	P1
Quick Links	Shortcuts to Manage Interns, Upload Tutorial, Generate Certificate, and Post Announcement.	P0

Module C: Attendance Management System

Feature	Description	Priority
Daily Check-In	Intern clicks Check-In button. System records Firestore timestamp and marks status 'present'.	P0
Daily Check-Out	Intern clicks Check-Out button. System records checkout timestamp. Duration is calculated.	P0
Duplicate Prevention	If intern has already checked in today, the Check-In button is disabled. Prevents double entries.	P0
Attendance History Table	Intern sees a paginated table of all past attendance records: date, check-in, check-out, duration.	P0
Attendance Percentage	Calculated as (days present / total working days) * 100. Displayed as a progress ring.	P0
Admin Monitoring View	Admin sees all interns' attendance in a filterable/sortable table with export to CSV.	P0
Team Lead View	Team Lead sees attendance only for their assigned team members.	P1
Late Arrival Flag	If check-in occurs after 10:00 AM, record is flagged as 'Late'. Configurable threshold.	P2
Monthly Summary	Admin can view and export a month-wise attendance summary per intern.	P1

Data Flow:

1. Intern clicks 'Check-In' on the dashboard.
2. Frontend calls Firestore and writes: { uid, date, checkIn: serverTimestamp(), status: 'present' }

3. Firestore Security Rules verify: the document uid matches the authenticated user.
4. On 'Check-Out', the same document is updated: { checkOut: serverTimestamp(), duration: calculated }
5. Admin reads /attendance collection with a composite query on date + status.

Module D: Tutorial & Learning Management System

A structured learning hub where admins upload resources by category and interns track their learning progress.

Feature	Description	Priority
Upload PDF Tutorial	Admin uploads a PDF via Firebase Storage. URL stored in Firestore. Interns view in-browser.	P0
Embed YouTube Video	Admin pastes a YouTube URL. System auto-generates an embedded player card.	P0
Tutorial Categories	Tutorials are organized into 6 categories: Power BI, Excel, SQL, Web Development, Design, AI Tools.	P0
Mark as Completed	Intern clicks 'Mark Complete' on any tutorial. Progress is stored in Firestore per user.	P0
Tutorial Progress Tracker	Visual progress bar per category showing X/Y tutorials completed.	P1
Tutorial Search	Full-text search within tutorial titles and descriptions.	P2
Admin Tutorial Manager	Admin can edit, delete, or reorder tutorials. Drag-and-drop ordering is a V2 feature.	P1
Team Lead Upload	Team Lead can upload tutorials; status is 'pending' until Admin publishes.	P2

Tutorial Categories — V1 Scope:

Power BI	Excel	SQL	Web Dev	Design	AI Tools
Data visualization	Spreadsheets	Databases	HTML/CSS/JS	Figma/Canva	ChatGPT, etc.

Module E: Offer Letter Generation System

Auto-generates professionally branded offer letters as downloadable PDFs using dynamic intern data.

Feature	Description	Priority
Auto-Generate Offer Letter	Admin selects an intern from a dropdown; system auto-fills all fields from Firestore.	P0
Startup Branding	Offer letter includes startup logo, name, registered address, and signature block.	P0
Dynamic Fields	Intern name, role, department, start date, end date, stipend, team lead name.	P0

PDF Download	Generated as a PDF using jsPDF or react-pdf. Downloadable by both Admin and Intern.	P0
Unique Letter ID	Each offer letter has a unique document ID (e.g., OL-2025-0034) stored in Firestore.	P1
Re-generate	Admin can regenerate a letter if intern data changes. Old version is archived.	P2

Module F: Certificate Generation System

Feature	Description	Priority
Internship Completion Certificate	Standard certificate issued at end of internship. Includes intern name, role, duration.	P0
Appreciation Certificate	Optional certificate for outstanding performance. Admin-discretion.	P1
QR Code Verification	Each certificate has a unique ID and a QR code linking to a public verification URL.	P0
PDF Download	Downloadable by the intern once approved. Admin can download any certificate.	P0
Admin Approval Workflow	Certificate is created in 'pending' state. Admin reviews and approves/rejects.	P0
Certificate ID	Format: CERT-2025-{internUID:6chars}. Unique, human-readable, verifiable.	P0
Revoke Certificate	Admin can revoke an approved certificate. Intern loses download access.	P2

Certificate Approval Workflow:

1. Admin navigates to Certificate Management in the Admin Panel.
2. Selects intern and certificate type (Completion or Appreciation).
3. Reviews auto-filled details: name, role, dates, duration.
4. Clicks 'Generate Preview' — PDF preview renders in a modal.
5. Admin clicks 'Approve & Publish'. Status updates to 'approved' in Firestore.
6. Intern receives an in-app notification and can now download the certificate.

Module G: Admin Panel

Feature	Description	Priority
Manage Interns	Full list of interns with filter/search. CRUD operations: add, edit, deactivate, delete.	P0
Monitor Attendance	Real-time attendance dashboard with date filters and CSV export.	P0

Upload Tutorials	Form to upload PDF or enter YouTube URL, select category, add title and description.	P0
Generate Documents	Unified interface for offer letter and certificate generation.	P0
Manage Users	Create/edit Team Lead accounts, assign interns to teams, reset passwords.	P0
Manage Announcements	Post, edit, and delete system-wide or team-scoped announcements.	P0
System Settings	Upload company logo, set company name, configure attendance thresholds.	P1
Export Data	Export attendance or intern list to CSV from any data table.	P1

4. Complete User Flow

4.1 Onboarding Flow

7. Admin creates a new intern account: name, email, department, team, start/end date.
8. System auto-generates a temporary password and sends it via Firebase Auth email.
9. Intern receives the email with login credentials and the platform URL.
10. On first login, intern is prompted to change their password.
11. Intern is redirected to the Intern Dashboard. A welcome announcement greets them.
12. Intern completes their profile: phone number, college name, avatar upload.

4.2 Login Flow

13. User visits the platform URL. The Auth Guard checks Firebase auth state.
14. If not authenticated, user is redirected to the Login page.
15. User enters email and password. Firebase Auth validates credentials.
16. On success, system reads `/users/{uid}` from Firestore to determine role.
17. User is routed to their role-specific dashboard: `/admin`, `/lead`, or `/intern`.
18. If authentication fails, an error message is shown. After 5 failed attempts, account is temporarily locked (Firebase default).

4.3 Attendance Flow

19. Intern logs in and sees the dashboard. The attendance card shows today's status.
20. If not checked in, 'Check-In' button is active. Intern clicks it.
21. System writes attendance record to Firestore with the server timestamp.
22. Dashboard updates in real-time: shows check-in time, button changes to 'Check-Out'.
23. Intern clicks 'Check-Out' at end of session. Record updated with checkout time and session duration.
24. Admin dashboard reflects updated attendance statistics instantly via Firestore real-time listener.

4.4 Tutorial Access Flow

25. Intern navigates to the Tutorials section via the sidebar.
26. Tutorials are displayed in category tabs (Power BI, Excel, SQL, etc.).
27. Intern selects a tutorial card — PDF opens in a new browser tab or embedded viewer; YouTube plays inline.
28. After consuming the tutorial, intern clicks 'Mark as Completed'.
29. Firestore updates the intern's progress document: `{ tutorialId: completed: true, completedAt: timestamp }`.
30. The category progress bar updates automatically.

4.5 Certificate Generation Flow

31. Admin opens Admin Panel > Certificate Management.

32. Selects intern from the list (only interns past their end date are eligible).
33. Chooses certificate type: Completion or Appreciation.
34. System auto-fills the certificate template: name, role, dates, QR code with unique CERT ID.
35. Admin previews the PDF in-browser, makes any edits to the description field.
36. Admin clicks 'Approve & Issue'. Firestore certificate document status is updated to 'approved'.
37. Intern sees a notification badge on their dashboard. Downloads the certificate PDF.

5. Database Design (Firestore)

The IMS uses Cloud Firestore (NoSQL document database) as its primary data store. All collections follow a flat structure optimized for Firestore's query model. Subcollections are used sparingly to avoid deeply nested paths.

5.1 Collection: /users

One document per registered user. The uid is the Firebase Auth UID (document ID).

Field	Type	Description	Required
Field	Type	Description	Required
uid	string	Firebase Auth UID — also the Firestore document ID	Yes
name	string	Full name of the user	Yes
email	string	Registered email address	Yes
role	string	Enum: 'admin' 'teamlead' 'intern'	Yes
department	string	Assigned department (e.g., 'Data Analytics')	Yes
teamLeadId	string	UID of the Team Lead assigned to this intern	No
startDate	timestamp	Internship start date	Yes
endDate	timestamp	Internship end date	Yes
phone	string	Mobile number (optional, added on first login)	No
college	string	College or university name	No
avatarUrl	string	Firebase Storage URL for profile photo	No
isActive	boolean	false if account has been deactivated by Admin	Yes
createdAt	timestamp	Account creation timestamp	Yes

5.2 Collection: /attendance

One document per intern per day. Document ID format: {uid}_{YYYY-MM-DD}

Field	Type	Description	Required
Field	Type	Description	Required
uid	string	Firebase Auth UID of the intern	Yes
date	string	Date in YYYY-MM-DD format for easy querying	Yes
checkIn	timestamp	Server-generated check-in timestamp	Yes

checkOut	timestamp	Server-generated check-out timestamp	No
durationMinutes	number	Calculated duration in minutes after check-out	No
status	string	Enum: 'present' 'late' 'absent'	Yes
isLate	boolean	true if checkIn time is after the configured late threshold	Yes

5.3 Collection: /tutorials

One document per tutorial resource uploaded by Admin or Team Lead.

Field	Type	Description	Required
Field	Type	Description	Required
id	string	Auto-generated Firestore document ID	Yes
title	string	Tutorial title	Yes
description	string	Short description (max 200 characters)	Yes
category	string	Enum: 'powerbi' 'excel' 'sql' 'webdev' 'design' 'aitools'	Yes
type	string	Enum: 'pdf' 'youtube'	Yes
url	string	Firebase Storage URL (PDF) or YouTube embed URL (video)	Yes
uploadedBy	string	UID of the uploader	Yes
isPublished	boolean	false until Admin approves (for Team Lead uploads)	Yes
order	number	Integer for manual display ordering within a category	No
createdAt	timestamp	Upload timestamp	Yes

5.4 Collection: /progress

One document per intern. Stores all tutorial completion status. Document ID = intern UID.

Field	Type	Description	Required
Field	Type	Description	Required
uid	string	Intern's Firebase Auth UID	Yes
completedTutorials	array<string>	Array of tutorial IDs the intern has marked complete	Yes
lastUpdated	timestamp	Timestamp of the last completion event	No

5.5 Collection: /offer_letters

Field	Type	Description	Required
Field	Type	Description	Required
id	string	Human-readable ID, e.g., OL-2025-0034	Yes
internUid	string	UID of the intern this letter belongs to	Yes
internName	string	Intern's full name at time of generation	Yes
role	string	Internship role/title	Yes
department	string	Department name	Yes
startDate	timestamp	Internship start date	Yes
endDate	timestamp	Internship end date	Yes
stipend	string	Monthly stipend amount (string to handle '0' / 'Unpaid')	No
generatedBy	string	UID of the Admin who generated the letter	Yes
pdfUrl	string	Firebase Storage URL of the generated PDF	Yes
createdAt	timestamp	Generation timestamp	Yes

5.6 Collection: /certificates

Field	Type	Description	Required
Field	Type	Description	Required
certId	string	Unique ID, format: CERT-2025-{6chars}	Yes
internUid	string	UID of the intern	Yes
internName	string	Intern's full name	Yes
type	string	Enum: 'completion' 'appreciation'	Yes
role	string	Internship role	Yes
department	string	Department name	Yes
startDate	timestamp	Internship start date	Yes
endDate	timestamp	Internship end date	Yes
status	string	Enum: 'pending' 'approved' 'revoked'	Yes
approvedBy	string	UID of the Admin who approved	No

approvedAt	timestamp	Approval timestamp	No
pdfUrl	string	Firebase Storage URL of the generated certificate PDF	No
qrData	string	URL encoded in the QR code for public verification	Yes
createdAt	timestamp	Request creation timestamp	Yes

5.7 Collection: /announcements

Field	Type	Description	Required
Field	Type	Description	Required
id	string	Auto-generated Firestore document ID	Yes
title	string	Announcement headline	Yes
body	string	Full announcement text (supports markdown)	Yes
postedBy	string	UID of the poster (Admin or Team Lead)	Yes
scope	string	Enum: 'global' 'team:{teamLeadId}'	Yes
createdAt	timestamp	Post timestamp	Yes
isPinned	boolean	If true, pinned to top of announcements feed	No

6. Recommended Tech Stack

Every technology below was selected on four criteria: zero or near-zero cost at startup scale, minimal DevOps overhead, proven reliability, and developer familiarity among intern-age developers.

Layer	Technology	Purpose	Why It Fits
Frontend	React 18	UI component library with hooks-based state management	Massive ecosystem; CRA/Vite scaffolding lets an intern-level dev be productive in day 1
Styling	Tailwind CSS 3	Utility-first CSS framework for rapid, responsive UI	No CSS files to maintain; design consistency enforced via config; dark mode trivial to add
Auth	Firebase Auth	Authentication provider with email, Google OAuth, and JWT tokens	Zero backend code; handles password reset, session management, and security automatically
Database	Firestore	NoSQL real-time document database	Real-time listeners eliminate polling; free Spark plan covers MVP; scales to millions of reads
File Storage	Firebase Storage	Cloud storage for PDFs, avatars, and certificate files	Same Firebase project; fine-grained security rules; CDN delivery built in
Hosting	Vercel	Frontend deployment and preview environments	Zero-config deployment from GitHub; preview URLs on every PR; free tier is sufficient
PDF Gen	jsPDF + html2canvas	Client-side PDF generation from HTML templates	No server needed; runs in the browser; intern-friendly React-based templates
QR Codes	qrcode.js	QR code generation for certificate verification links	Zero-cost; 1KB library; generates SVG or canvas QR codes client-side
State Mgmt	Zustand	Lightweight global state management	Far simpler than Redux; single store file; works perfectly with Firebase real-time listeners
Charts	Recharts	React-native charting library for dashboard analytics	Built for React; simple API; covers bar, line, donut charts needed for dashboards

Cost at MVP Scale: Firebase Spark (free) plan supports 50K reads/day, 20K writes/day, 1GB storage.

For 20-25 interns, estimated usage is well under 1% of free quota. Vercel free tier supports unlimited deployments. Total infrastructure cost at MVP scale: \$0/month.

7. UI/UX Recommendations

7.1 Layout Architecture

- Fixed left sidebar (240px wide on desktop, collapsible to 64px icon-only mode).
- Top header bar showing current page title, user avatar, and notification bell.
- Main content area in a max-width-7xl centered container with 24px padding.
- Responsive: sidebar collapses to a bottom nav bar on mobile (screen < 768px).

7.2 Design System

Token	Value	Usage
Primary	#1E3A5F	Sidebar bg, headings, primary buttons
Accent	#2D7DD2	Links, highlights, active states
Background	#F8FAFC	Page background
Card	#FFFFFF	Dashboard cards, modal backgrounds
Border	#E2E8F0	Card borders, dividers
Success	#1A7340	Attendance present, completion badges
Warning	#B45309	Late arrival, pending status
Error	#B91C1C	Absent, rejected, error states
Font — Body	Inter, 14px	All body text and labels
Font — Mono	JetBrains Mono	Code snippets, IDs, timestamps
Border Radius	12px cards, 8px buttons	Consistent rounded corners

7.3 Component Patterns

- Stat Cards: Icon + metric + label + subtle trend arrow. Used on all dashboard pages.
- Data Tables: Sortable columns, row hover highlight, search bar above, pagination below.
- Modal Dialogs: Centered overlay for form actions (add intern, generate certificate). Max width 640px.
- Toast Notifications: Bottom-right corner, auto-dismiss after 4 seconds. Success/error variants.
- Empty States: Illustrated empty state for new interns (no tutorials yet, no attendance records).
- Loading States: Skeleton loaders (not spinners) for all data-fetching components.

7.4 Accessibility & Responsiveness

- All interactive elements have ARIA labels and keyboard focus states.
- Minimum touch target 44x44px on mobile for all buttons and links.
- Color contrast ratio $\geq 4.5:1$ for all body text (WCAG AA compliance).
- PDF certificates and offer letters are generated in A4 format for universal printing.

8. MVP Scope (V1)

8.1 Include in V1

Feature	Rationale	Priority
Email/password authentication + RBAC	Core functionality; everything depends on it	P0 — Must Have
Admin Dashboard with key stats	Gives admins immediate value on day 1	P0 — Must Have
Intern Dashboard with quick check-in	Core daily interaction for interns	P0 — Must Have
Attendance check-in/check-out	Primary pain point being solved	P0 — Must Have
Attendance history view	Interns need visibility into their records	P0 — Must Have
Tutorial upload (PDF + YouTube)	Learning is a key internship value-add	P0 — Must Have
Tutorial completion tracking	Progress visibility for interns	P0 — Must Have
Offer letter PDF generation	Replaces manual Word document process	P0 — Must Have
Certificate generation + approval	Critical for intern program completion	P0 — Must Have
Announcements system	Replaces WhatsApp broadcast for comms	P1 — Should Have
Admin: Manage intern accounts	Admin needs CRUD for intern management	P0 — Must Have
CSV export for attendance	Admin operational need for records	P1 — Should Have

8.2 Do NOT Build in V1

- Task Management System: Adds significant complexity; out of scope for internship MVP.
- AI Analytics / Insights: Save for V2 once data volume justifies it.
- Leaderboard / Gamification: Nice-to-have; risks intern comparison anxiety.
- Mobile Native App: The responsive web app covers mobile needs for MVP.
- Bulk CSV Import of Interns: Manual Admin creation is fine for 20-25 interns.
- Multi-tenant / Multi-company: Focus on single-startup deployment for V1.
- Slack / Google Calendar Integration: Adds API complexity; email is sufficient for V1.
- Video Recording / Upskill Platform: Out of scope; YouTube embeds cover this.

9. Development Roadmap (3-Week MVP Sprint)

The roadmap assumes a team of 2-3 developers (1 senior + 1-2 junior/intern developers) working full time. Each week has a clear deliverable milestone that can be demoed to stakeholders.

Week	Phase	Deliverables	Goal
Week 1 Days 1-7	Foundation & Auth	Project setup (Vite + React + Tailwind), Firebase project config, Auth system (login/signup/route guards), Firestore schema setup, Sidebar layout + navigation shell, Intern Dashboard skeleton, Admin Dashboard skeleton	Working auth flow; role-based routing; empty dashboards deployed to Vercel
Week 1 Days 5-7	Attendance Core	Check-in / Check-out UI + Firestore writes, Attendance history table for interns, Admin attendance monitoring view, Attendance percentage calculation, Firestore Security Rules v1	End-to-end attendance flow working; intern can check in and see history
Week 2 Days 8-11	Tutorial LMS	Tutorial upload form (PDF + YouTube URL), Firebase Storage integration, Tutorial card grid with category tabs, Mark-as-Complete functionality, Progress bar components	Admin can upload tutorials; interns can browse and track completion
Week 2 Days 11-14	Document Generation	Offer letter HTML template + jsPDF generation, Certificate HTML template + QR code, Admin approval workflow UI, Firestore document status management, PDF download + Firebase Storage upload	Admin can generate offer letters and certificates; interns can download theirs
Week 3 Days 15-18	Admin Panel & Polish	Manage Interns CRUD, Announcements system, CSV export for attendance, System Settings (logo, company name), Toast notifications, Empty states + skeleton loaders	Full Admin Panel functional; announcements working; polished UX
Week 3 Days 19-21	Testing & Deployment	End-to-end testing of all user flows, Firestore Security Rules audit, Performance optimization (lazy loading), Custom domain setup on Vercel, Onboarding documentation for Admin, Staging + Production environments	Production-ready deployment; Admin onboarding guide delivered

9.1 Daily Standup Cadence

- 15-minute async standup via shared doc: What did I do yesterday? What will I do today? Any blockers?

- Weekly demo every Friday: Working feature demo to stakeholders, no slides required.
- GitHub PR reviews within 4 hours for all team members.

10. Project Architecture & Folder Structure

10.1 Recommended Folder Structure

```

ims/
├── public/
│   └── favicon.ico
├── src/
│   ├── assets/           # Static images, fonts, logo
│   ├── components/       # Reusable UI components
│   │   ├── common/       # Button, Input, Modal, Badge, Skeleton
│   │   ├── layout/       # Sidebar, Header, Footer, PageWrapper
│   │   ├── dashboard/    # StatCard, AttendanceCard, TutorialProgress
│   │   ├── attendance/   # CheckInButton, AttendanceTable, AttendanceChart
│   │   ├── tutorials/    # TutorialCard, CategoryTabs, ProgressBar
│   │   ├── documents/    # OfferLetterTemplate, CertificateTemplate
│   │   └── admin/        # InternTable, ApprovalModal, SettingsForm
│   ├── pages/            # Route-level page components
│   │   ├── auth/         # LoginPage, ForgotPasswordPage
│   │   ├── intern/       # InternDashboard, MyAttendance, Tutorials
│   │   ├── admin/        # AdminDashboard, ManageInterns, AdminTutorials
│   │   │   ├── documents/ # OfferLetters, Certificates
│   │   │   └── settings/  # SystemSettings
│   │   └── lead/         # LeadDashboard, TeamAttendance
│   ├── hooks/            # Custom React hooks
│   │   ├── useAuth.js    # Auth state + role from Firestore
│   │   ├── useAttendance.js # Attendance CRUD + real-time listener
│   │   ├── useTutorials.js # Tutorials fetch + progress tracking
│   │   └── useInterns.js  # Intern list management (admin only)
│   ├── store/            # Zustand global state stores
│   │   ├── authStore.js  # User object, role, loading state
│   │   └── uiStore.js    # Sidebar open/close, toast queue
│   ├── services/         # Firebase abstraction layer
│   │   ├── firebase.js   # Firebase app initialization
│   │   ├── authService.js # login(), logout(), resetPassword()
│   │   ├── userService.js # getUser(), createUser(), updateUser()
│   │   ├── attendanceService.js
│   │   ├── tutorialService.js
│   │   ├── certificateService.js
│   │   └── storageService.js
│   ├── utils/            # Pure utility functions
│   │   ├── dateUtils.js  # formatDate(), calcDuration(), isLate()
│   │   ├── pdfUtils.js   # generateOfferLetterPDF(), generateCertPDF()
│   │   └── qrUtils.js     # generateQRCode(certId)
│   ├── constants/        # App-wide constants
│   │   ├── roles.js      # ADMIN, TEAM_LEAD, INTERN constants
│   │   ├── categories.js # Tutorial category definitions
│   │   └── routes.js     # All route path constants
│   ├── App.jsx           # Root component with React Router
│   └── main.jsx           # Vite entry point
├── .env                  # Firebase config (never commit to git)
├── .env.example          # Template for .env variables
├── firestore.rules       # Firestore Security Rules
├── storage.rules         # Firebase Storage Security Rules
├── vite.config.js
└── package.json

```


10.2 Firebase Security Rules (Firestore)

```

rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {

    // Helper: check if authenticated
    function isAuth() { return request.auth != null; }

    // Helper: get the caller's role from Firestore
    function getRole() {
      return
get(/databases/{database}/documents/users/{request.auth.uid}).data.role;
    }

    // Users collection
    match /users/{uid} {
      allow read: if isAuth() && (request.auth.uid == uid || getRole() ==
'admin');
      allow create: if isAuth() && getRole() == 'admin';
      allow update: if isAuth() && (request.auth.uid == uid || getRole() ==
'admin');
      allow delete: if isAuth() && getRole() == 'admin';
    }

    // Attendance collection
    match /attendance/{docId} {
      allow read: if isAuth();
      allow create: if isAuth() && request.resource.data.uid == request.auth.uid;
      allow update: if isAuth() && resource.data.uid == request.auth.uid;
    }

    // Tutorials - anyone can read published; only admin/lead can write
    match /tutorials/{tutId} {
      allow read: if isAuth() && resource.data.isPublished == true;
      allow write: if isAuth() && (getRole() == 'admin' || getRole() ==
'teamlead');
    }

    // Certificates - admin writes; intern reads own
    match /certificates/{certId} {
      allow read: if isAuth() && (resource.data.internUid == request.auth.uid
|| getRole() == 'admin');
      allow write: if isAuth() && getRole() == 'admin';
    }
  }
}

```

10.3 PDF Generation Workflow

38. Admin fills out the document generation form (intern name, dates, etc.).
39. React renders an off-screen HTML component styled to A4 dimensions (794 x 1123px).
40. html2canvas captures the rendered HTML as a canvas image at 2x device pixel ratio for crisp output.
41. jsPDF creates a new PDF document and adds the canvas image as the page content.
42. The PDF blob is uploaded to Firebase Storage at path: /documents/{type}/{certId}.pdf.

43. The download URL is stored in the Firestore document. Intern accesses via their dashboard.

PDF Template Tip: Build your offer letter and certificate as React components with inline styles (Tailwind's JIT classes may not be captured by html2canvas). Use exact px dimensions for A4: width: 794px, height: 1123px. Test the PDF output in Chrome, Firefox, and Safari before shipping.

11. Future Scope (V2 & Beyond)

The IMS is architected to grow. These features are explicitly out of V1 scope but are designed-for in the data model and architecture to minimize future migration effort.

Feature	Description	Target Version
Task Management	Kanban board for assigning tasks to interns. Track status: To Do, In Progress, Done.	V2
Leaderboard	Optional gamified attendance and tutorial leaderboard. Admin can toggle on/off.	V2
AI Writing Assistant	In-app AI assistant (via Claude API) to help interns write reports and summaries.	V2
Internship Analytics	Advanced cohort analytics: dropout rate, avg attendance %, top categories, retention.	V2
Notification System	In-app + email notifications for certificate approvals, announcements, and reminders.	V2
Multi-Cohort Support	Manage multiple internship cohorts simultaneously (e.g., Summer 2025, Winter 2025).	V3
Mobile App (PWA)	Progressive Web App installable on Android and iOS from the browser.	V3
AI Attendance Anomaly Detection	Flag unusual patterns: intern who never checks out, sudden absence streaks.	V3
Intern Portfolio Generator	Auto-generate a shareable internship portfolio PDF from tutorials and tasks completed.	V3
External Certificate Verifier	Public web page where employers can scan the QR code to verify a certificate.	V1.5
Bulk Intern CSV Import	Admin can import 20+ intern accounts from a CSV file in a single operation.	V2
Google Calendar Sync	Sync internship schedule and check-in reminders to intern Google Calendar.	V3

12. Deployment Recommendations

12.1 Environments

Environment	Description
Development	Local machine. Firebase emulators for Auth and Firestore (firebase emulators:start). No real data.
Staging	Vercel preview deployment auto-created on every GitHub PR. Separate Firebase project.
Production	Vercel production deployment from main branch. Primary Firebase project with real intern data.

12.2 Environment Variables (.env)

```
VITE_FIREBASE_API_KEY=your_api_key
VITE_FIREBASE_AUTH_DOMAIN=your_project.firebaseio.com
VITE_FIREBASE_PROJECT_ID=your_project_id
VITE_FIREBASE_STORAGE_BUCKET=your_project.appspot.com
VITE_FIREBASE_MESSAGING_SENDER_ID=your_sender_id
VITE_FIREBASE_APP_ID=your_app_id
VITE_APP_NAME=Internship Management System
VITE_COMPANY_NAME=Your Startup Name
VITE_CERT_VERIFY_BASE_URL=https://your-domain.com/verify
```

12.3 Deployment Checklist

- Firestore Security Rules deployed and tested with the Rules Playground.
- Firebase Storage rules restrict uploads to authenticated users.
- Environment variables set in Vercel Dashboard (Settings > Environment Variables).
- Custom domain configured in Vercel with SSL auto-provisioned.
- Firebase project billing alerts set at \$1, \$5, and \$10 thresholds.
- Google Analytics 4 connected to Firebase for usage monitoring.
- Admin account created first before any intern onboarding begins.

Appendix A: Glossary

Term	Definition
IMS	Internship Management System — the platform described in this PRD
RBAC	Role-Based Access Control — permissions system based on assigned role
Firestore	Google Cloud Firestore — the NoSQL document database used by the IMS
JWT	JSON Web Token — Firebase Auth issues JWTs for session authentication
PRD	Product Requirements Document — this document
MVP	Minimum Viable Product — the V1 scope described in Section 8
P0 / P1 / P2	Priority levels: P0 = must have for MVP, P1 = should have, P2 = nice to have
QR Code	Quick Response code embedded in certificates for external verification
jsPDF	JavaScript library for generating PDF documents client-side in the browser
Vercel	Frontend hosting and deployment platform used for the IMS web application
Cohort	A group of interns enrolled in the same internship program cycle

Appendix B: Document Control

Attribute	Value
Document Title	Internship Management System — Product Requirements Document
Version	1.0 — MVP Edition
Status	Draft — Pending Stakeholder Review
Author	Product Team
Created	May 2025
Last Updated	May 2025
Review Cycle	Every sprint (2 weeks)
Distribution	Internal — Engineering, Design, Product
Next Review	End of Week 1 of development sprint

— End of Document —